The Time Capsule Switcheroo

Whodunnit? · Exponential & Logarithmic Functions · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Exponential & Logarithmic Functions	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Evaluating exponential and log expressions, log properties, change of base, solving exponential and log equations (with extraneous roots), compound and continuous growth.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Ezra Lindgren took it, in the Bandstand, using this method: Tucked in a hollow book.

The reveal (read aloud): While the crowd cheered the ribbon-cutting, **Ezra Lindgren** strolled over to the **Bandstand**, slipped the Founders' Pocket Watch inside a **hollow book** from the museum's gift table, and tucked it under his arm. Every alibi and method you cleared pointed straight at him. Time well spent, detective!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Dr. Imani Okoye · Place: Museum Annex
2	Method: Hidden in a top hat
3	Suspect: Mayor Renata Voss · Place: Town Hall Steps
4	Method: Swapped for a replica
5	Suspect: Priya Nambiar · Place: Gazette Office
6	Method: Sewn into the banner
7	Suspect: Tobias Quill · Place: Clock Tower
8	Method: Stashed in the scissors case
9	Suspect: Dottie Marsh · Place: Bank Vault
10	Method: Buried in a flower bouquet

Answer Key

The Time Capsule Switcheroo · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Find the horizontal asymptote: $y = 2(\frac{1}{2})^{x+1} - 3$	**B.** $y = -3$	The base is $\frac{1}{2}$, so this is exponential decay. As x grows, the power $(\frac{1}{2})^{x+1}$ gets closer and closer to 0. So y gets closer and closer to $2(0) - 3 = -3$. The horizontal asymptote is the line $y = -3$	Dr. Okoye was cataloguing donated letters in the Museum Annex. The sign-in sheet has her handwriting on every page.
2	Evaluate without a calculator: $\log_8 4$	**D.** $\frac{2}{3}$	Let $\log_8 4 = y$. In exponential form, $8^y = 4$. Write both sides as powers of 2: $(2^3)^y = 2^2$, so $2^{3y} = 2^2$. $3y = 2$, so $y = \frac{2}{3}$	Every top hat in the parade was checked at the gate. The only thing under them was hair.
3	Solve for x: $\log_3 x = -2$	**A.** $x = \frac{1}{9}$	Rewrite in exponential form: $x = 3^{-2}$ A negative exponent means reciprocal: $3^{-2} = \frac{1}{3^2}$ $x = \frac{1}{9}$	Mayor Voss was cutting the ribbon on the Town Hall Steps in front of two hundred cameras.
4	A deposit of $2,000 earns 4% interest compounded quarterly. Using $A = P(1 + \frac{r}{n})^{nt}$, find the balance after 5 years, to the nearest cent.	**B.** $2,440.38	$P = 2000$, $r = 0.04$, $n = 4$, $t = 5$ $A = 2000(1 + \frac{0.04}{4})^{4 \cdot 5} = 2000(1.01)^{20}$ Do not round until the end: $2000 \times 1.2201900... \approx$ $2,440.38	The velvet box held its original tag. The Watch was not swapped for a replica; the box was simply empty.
5	Evaluate: $\log_3 6 + \log_3 9 - \log_3 2$	**C.** 3	Product and quotient properties: add the logs of the numerators, subtract the log of the denominator. $\log_3 \frac{6 \cdot 9}{2} = \log_3 27$ $3^3 = 27$, so the value is 3	Priya Nambiar was filing her story from the Gazette Office. The server stamped her upload at the exact minute of the theft.
6	Evaluate to the nearest thousandth: $\log_3 20$	**D.** 2.727	Change of base: $\log_3 20 = \frac{\ln 20}{\ln 3}$ $\approx \frac{2.9957}{1.0986}$ ≈ 2.727	The centennial banner was unstitched and shaken out. Nothing was sewn inside.

7	Solve for x: $4^{x+1} = 8^{x-1}$	**C.** $x = 5$	Write both sides with base 2: $(2^2)^{x+1} = (2^3)^{x-1}$ $2^{2x+2} = 2^{3x-3}$, so the exponents are equal: $2x + 2 = 3x - 3$ $x = 5$	Tobias Quill was repairing the big clock in the Clock Tower. The gears were still oily when the police arrived.
8	Solve for x to the nearest thousandth: $6 \cdot 2^x = 150$	**B.** $x \approx 4.644$	Isolate the power first: divide both sides by 6, so $2^x = 25$ Take the natural log of both sides: $x \cdot \ln 2 = \ln 25$ $x = \frac{\ln 25}{\ln 2} \approx 4.644$	The ribbon-cutting scissors case was opened by the mayor’s aide. It held only scissors.
9	Solve for x and check for extraneous solutions: $\log_2 x + \log_2(x - 2) = 3$	**A.** $x = 4$	Product property: $\log_2[x(x - 2)] = 3$, so $x(x - 2) = 2^3 = 8$ $x^2 - 2x - 8 = 0$, so $(x - 4)(x + 2) = 0$, giving $x = 4$ or $x = -2$ Check $x = -2$: $\log_2(-2)$ is undefined, so -2 is extraneous. Check $x = 4$: $\log_2 4 + \log_2 2 = 2 + 1 = 3$ ✓ $x = 4$	Dottie Marsh was counting centennial coins in the Bank Vault. The vault door log shows she never stepped out.
10	$5,000 is invested at 6% interest compounded continuously, using $A = Pe^{rt}$. How many years will it take to double, to the nearest tenth of a year?	**C.** 11.6 years	Set $A = 10{,}000$: $10{,}000 = 5000e^{0.06t}$, so $e^{0.06t} = 2$ Take the natural log of both sides: $0.06t = \ln 2$ $t = \frac{\ln 2}{0.06} \approx 11.55$, which rounds to 11.6	Every bouquet was sniffed, shaken and pulled apart. Not a single watch chain in the roses.

Wrong-Answer Map

The Time Capsule Switcheroo · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $x = -3$	wrote a vertical line; a horizontal asymptote has the form $y = k$	The museum gift table was searched and every book was a solid hardcover. Nothing hollow, nothing hidden.
1	C. $y = 3$	sign error on the vertical shift	Ezra Lindgren was dusting display cases in the Museum Annex all morning. Three volunteers say he never left.
1	D. $y = 0$	forgot the vertical shift; used the asymptote of the parent graph	Tobias Quill was tuning the carillon controls at the Bandstand. A trumpet player videoed him the entire time.
2	A. $\frac{1}{2}$	divided 4 by 8 instead of solving $8^y = 4$	Dr. Imani Okoye was reading the town charter aloud at the Bandstand. The microphone log has her voice for the full hour.
2	B. $\frac{3}{2}$	inverted the ratio: 8 and 4 are powers of 2, so the exponent is 2 over 3, not 3 over 2	Ezra Lindgren was climbing the Clock Tower stairs with a tour group. The guide counted him at every landing.
2	C. 2	stopped at $4 = 2^2$ and never adjusted for the base $8 = 2^3$	The library cart was inventoried item by item. No book had been cut open or hollowed out.
3	B. $x = 9$	dropped the negative sign of the exponent	Ezra Lindgren was holding the ribbon for the mayor on the Town Hall Steps. Every news photo shows him there.
3	C. $x = -9$	treated 3^{-2} as $-(3^2)$	Mayor Renata Voss was posing for portraits at the Bandstand until noon. The photographer's timestamps back her up.
3	D. $x = -6$	multiplied the base by the exponent instead of raising it	A gift-table volunteer insists no book left the museum table. Every title was accounted for at closing.
4	A. $4,382.25	forgot to divide the rate by 4 but still used 20 periods	Dottie Marsh was handing out commemorative coins at the Bandstand. A line of forty kids can vouch for her.
4	C. $2,102.02	used 1.01 but forgot to multiply the exponent by n (5 instead of 20)	Police shook out every book on the display shelf. Pages fluttered, but no watch fell out.
4	D. $2,433.31	compounded once a year instead of quarterly (n = 1)	Ezra Lindgren was handing the Gazette a quote in the Gazette Office. The editor logged the call at the time of the theft.
5	A. $\log_3 108$	multiplied by 2 instead of dividing: the quotient property subtracts	A hollow-book hunt came up empty: security flipped through every volume and found only ordinary pages.
5	B. $\log_3 13$	added and subtracted the numbers inside the logs: log of a sum is not a sum of logs	Priya Nambiar was interviewing the brass band at the Bandstand. Her recorder has the whole conversation.
5	D. 27	combined correctly but never took the log: $\log_3 27 = 3$	Ezra Lindgren was admiring the capsule's coins inside the Bank Vault. The vault camera shows him with the guards.
6	A. 6.667	divided 20 by 3; log base 3 is not division	Mayor Renata Voss was waiting to cut the cake at the Bandstand. A dozen councilors stood right beside her.
6	B. 1.301	found log 20 only (base 10) and ignored the base 3	The hollow-book idea is a dead end: the only book with a cavity was a prop in the window display, and it was empty.

6	C. 0.367	inverted the change-of-base ratio: ln 3 over ln 20	Ezra Lindgren was relabeling exhibits in the Museum Annex. The curator’s time card has him clocked in all morning.
7	A. $x = 4$	distributed the 3 but forgot to distribute the 2	The town’s book drive was weighed and checked. No book felt too light, and none had a hidden pocket.
7	B. $x = -1$	sign error when distributing: $2x + 2 = 3x + 3$	Ezra Lindgren was winding the Clock Tower’s antique pendulum with Mr. Quill. Both signed the maintenance log.
7	D. $x = 3$	used the bases 4 and 8 as multipliers instead of rewriting with a common base	Dottie Marsh was announcing raffle winners at the Bandstand. The microphone log shows she never left the stage.
8	A. $x \approx 2.016$	multiplied 6 and 2 to get 12^x, which is not allowed	Security X-rayed the museum’s book crate at the gate. Every book was solid paper and nothing metal showed up.
8	C. $x \approx 7.229$	took the log of 150 without dividing by 6 first	Ezra Lindgren was introducing the speakers on the Town Hall Steps. The livestream shows him the entire hour.
8	D. $x \approx 1.609$	divided ln 25 by the base 2 instead of by ln 2	Dr. Imani Okoye was signing history pamphlets at the Bandstand. A long line of readers waited with her.
9	B. $x = 4$ or $x = -2$	did not check for extraneous solutions: -2 makes a log argument negative	Tobias Quill was testing the speakers at the Bandstand. The sound engineer recorded him at the mixing board.
9	C. $x = -2$	chose the wrong root: $\log_2(-2)$ is undefined	Ezra Lindgren was proofreading the commemorative edition in the Gazette Office. The editor has his corrections in red pen.
9	D. $x = 3$	set $x(x - 2)$ equal to 3 instead of $2^3 = 8$	Every book on the donation table was shaken, flipped and measured. None was hollow.
10	A. 0.04 years	multiplied ln 2 by 0.06 instead of dividing	The hollow book on the gift table turned out to be a plastic prop with a sealed lid. It was sealed and empty.
10	B. 33.3 years	set $0.06t = 2$ and forgot to take the natural log	Priya Nambiar was photographing the band at the Bandstand. Her camera roll has them every thirty seconds.
10	D. 11.9 years	used ln 1.06 (annual compounding) instead of the rate 0.06 for continuous growth	Ezra Lindgren was signing the vault guestbook in the Bank Vault. Ms. Marsh’s pen is still on the page.